ARKADIP SOM

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Education

Indian Institute of Engineering Science and Technology, Shibpur **Aug 2024 – Present**
B.Tech in Civil Engineering CGPA: 7.88 (Till 3rd Semester)

Begampur High School (WBCHSE) **2021 – 2023**
Higher Secondary 87.8%

Begampur High School (WBBSE) **2016 – 2021**
Secondary (Madhyamik) 97.8%

Experience

Case Study **Jan 2026 – April 2026**
Bureau of Indian Standards (BIS)

- Evaluated quality maintenance practices per BIS cement standards **IS 269:2015**, **IS 1489 (Part 1 & 2):2015**, and **IS 455 (Part 1):2015**.
- Conducted industrial visit to **UltraTech Cement (Dankuni, Hooghly)** to observe quality control and standardization procedures in practice.
- Performed standard cement quality tests (fineness, setting time, strength) and analyzed results against BIS acceptance criteria to identify gaps between prescribed standards and on-ground implementation.

Projects

Kolkata Water Infrastructure Forecast Dashboard Live App | GitHub
Python · Streamlit · Scikit-Learn · Plotly · Pandas · NumPy *2026*

- Built a full-stack forecasting dashboard integrating **4 population models** (Arithmetic, Geometric, Linear & Polynomial Regression) with civil engineering demand norms to project Kolkata's water demand and wastewater load through 2050.
- Engineered a wastewater load distribution module mapping sewage generation across **12+ STPs**, computing real-time capacity stress (safe / near-capacity / overloaded) against IS-code parameters including BOD (250 mg/L), COD (500 mg/L), and peak flow factors.
- Simulated the complete urban water pipeline — population → demand → wastewater → sewer flow → STP capacity — with interactive Plotly visualisations for municipal infrastructure decision support.
- Deployed as a **live Streamlit Cloud application**, bridging ML-based data science with domain-specific civil engineering standards, demonstrating cross-disciplinary integration rare among undergraduate portfolios.

Weather Prediction in Data-Scarce Conditions GitHub
Python · Scikit-Learn *2026*

- Built ML regression models to predict weather variables from limited historical datasets, applying missing value imputation, normalization and feature engineering to handle data scarcity.
- Applied cross-validation to reduce overfitting; evaluated model robustness using RMSE and R^2 score under varying data availability.

House Price Prediction GitHub
Python · Scikit-Learn · Pandas · NumPy *2026*

- Built a supervised ML pipeline to predict residential house prices using regression models with feature engineering, data cleaning and normalization on structured tabular data.
- Evaluated and compared model performance using RMSE and R^2 score to select the best-fit estimator for price prediction.

Skills Summary

Languages: Python, C, C++
Libraries: NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch
Tools: Git, VS Code, Google Colab, Jupyter Notebook
Platforms: GitHub
Civil Engineering: AutoCAD